# Block practice · Series and the binomial expansion

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Expanding a bracket with a negative or fractional power

When is a binomial expansion valid?

Getting a bracket ready for the binomial series: a(1 + bx)^n

Finding the first term and common difference of an arithmetic series

Using a binomial expansion to estimate a value

Finding n from an equation linking sums and terms

Finding the first term and common ratio of a geometric series

Does a geometric series have a sum to infinity?

Sum to infinity of a geometric series

Adding up a finite geometric series

Finding n in a geometric series with logarithms

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# Practice workbook · Expanding a bracket with a negative or fractional power

Expanding (1 + kx)^n for rational n in ascending powers of x up to a stated term, with coefficients simplified.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-18

## 1. Question 9 · part a · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

**9** (a) Expand $\sqrt{1-4x}$ in ascending powers of $x$ up to and including the term in $x^3$, giving each coefficient as an integer.

**(3)**

(b) Use your expansion with a suitable value for $x$ to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

**(3)**

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

**(2)**

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## 1. Question 9 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 9 · part a · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part a only.

**9** (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of $x$ up to and including the term in $x^3$

Simplify each term as far as possible.

**(3)**

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

**(2)**

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where $a$, $b$ and $c$ are integers.

Show all your working clearly.

**(2)**

Using the expansion you found in part (a) with a suitable value of $x$

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

**(4)**

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## 2. Question 9 · part a · 3 marks · Working and answer

Working and answer

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## 3. Question 6 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

**6** Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where $p$ and $q$ are constants,

(a) find the value of $p$ and the value of $q$

**(2)**

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of $x$ up to and including the term in $x^2$, expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

Using the expansion found in part (b) with a suitable value of $x$

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

**(2)**

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## 3. Question 6 · part b · 3 marks · Working and answer

Working and answer

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## 4. Question 6 · part b · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

**6** (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where $A$ and $B$ are integers to be found.

**(2)**

(b) Hence expand $\sqrt{4-x}$ in ascending powers of $x$ up to and including the term in $x^3$
Give each coefficient as an exact fraction in its lowest terms.

**(3)**

(c) Use your expansion with a suitable value of $x$ to obtain an estimate of $\frac{\sqrt{385}}{10}$

Give your answer correct to 5 decimal places.

**(3)**

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## 4. Question 6 · part b · 3 marks · Working and answer

Working and answer

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## 5. Question 6 · part b · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**6** Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where $p$ and $q$ are constants

(a) find the value of $p$ and the value of $q$.

**(2)**

(b) Hence expand $\sqrt{9-x}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

Using the expansion you found in part (b) with a suitable value of $x$,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

**(3)**

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## 5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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## 6. Question 1 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

**1** (a) Expand $\left(1+\frac{x}{4}\right)^{8}$ in ascending powers of $x$ up to and including the term in $x^3$

Give each coefficient in its simplest terms.

**(3)**

(b) Use your expansion with a suitable value of $x$ to obtain an approximation, to 4 decimal places, of $(1.035)^8$

**(3)**

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**(Total for Question 1 is 6 marks)**

## 6. Question 1 · part a · 3 marks · Working and answer

Working and answer

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## 7. Question 3 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

**3** (a) Expand

$$(1 + px)^{-5} \quad p \neq 0$$

in ascending powers of $x$, up to and including the term in $x^4$
Give each term in its simplest form.

(3)

The coefficient of $x^r$ in the expansion is $c_r$

Given that $c_4 = 2c_3$

(b) find the value of $p$.

(2)

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## 7. Question 3 · part a · 3 marks · Working and answer

Working and answer

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## 8. Question 9 · part a · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

**9** (a) Expand $(1+2x)^{-\frac{1}{3}}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as a fraction in its lowest terms.

**(3)**

(b) Find the range of values of $x$ for which your expansion is valid.

**(1)**

$$f(x) = \frac{2+kx^2}{(1+2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f $(x)$ in ascending powers of $x$ up to and including the term in $x^3$
Give your coefficients in terms of $k$ where appropriate.

**(3)**

The coefficient of $x^3$ in the series expansion of f $(x)$ is $-\frac{8}{3}$

(d) Find the exact value of $k$

**(2)**

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

**(5)**

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## 8. Question 9 · part a · 3 marks · Working and answer

Working and answer

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## 9. Question 2 · part a · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

**2** (a) Expand $\frac{2}{\sqrt{1+3x}}$ in ascending powers of $x$ up to and including the term in $x^3$

Express each coefficient as a fraction in its simplest terms where appropriate.

**(4)**

(b) State the range of values of $x$ for which the expansion is valid.

**(1)**

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## 9. Question 2 · part a · 4 marks · Working and answer

Working and answer

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## 10. Question 1 · part whole · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

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**1** Expand $\left(1-\frac{x}{4}\right)^{-2}$ in ascending powers of $x$ up to and including the term in $x^3$

Where appropriate, express each coefficient as an exact fraction in its lowest terms.

**(3)**

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**(Total for Question 1 is 3 marks)**

## 10. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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## 11. Question 5 · part b · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

**5** Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where $p$ and $q$ are constants,

(a) find the value of $p$ and the value of $q$

**(2)**

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of $x$ up to and including the term in $x^3$, expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of $x$ up to and including the term in $x^3$, expressing each coefficient as an exact fraction in its lowest terms.

**(2)**

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

**(4)**

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## 11. Question 5 · part b · 3 marks · Working and answer

Working and answer

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## 12. Question 7 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

**7** (a) Expand $\left(1+2x^2\right)^{-\frac{3}{4}}$ in ascending powers of $x$ up to and including the term in $x^6$

Express each coefficient as an exact fraction in its lowest terms.

**(3)**

$$f(x) = \frac{(2+kx)}{\left(1+2x^2\right)^{\frac{3}{4}}} \quad \text{where } k \neq 0$$

(b) Obtain a series expansion for $f(x)$ in ascending powers of $x$ up to and including the term in $x^5$

Give each coefficient in terms of $k$ where appropriate.

**(2)**

The coefficient of the term in $x^5$ is fourteen times the coefficient of the term in $x^2$

(c) Find the value of $k$

**(2)**

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## 12. Question 7 · part a · 3 marks · Working and answer

Working and answer

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## 13. Question 9 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**9** Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of $p$ and the value of $q$.

**(2)**

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of $x$ up to and including the term in $x^3$ and express each coefficient as an exact fraction in its lowest terms.

**(3)**

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of $a$ and the value of $b$.

**(3)**

(d) Find the exact value of $c$.

**(2)**

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## 13. Question 9 · part b · 3 marks · Working and answer

Working and answer

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## 14. Question 9 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**9** (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where $a$ and $b$ are fractions in their lowest terms.

**(2)**

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of $x$ up to and including the term in $x^3$
Express each coefficient as a fraction in its lowest terms.

**(3)**

(c) (i) Use a suitable value of $x$ with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

**(4)**

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## 14. Question 9 · part b · 3 marks · Working and answer

Working and answer

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## 15. Question 10 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

**10** (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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## 15. Question 10 · part b · 3 marks · Working and answer

Working and answer

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## 16. Question 7 · part a · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

**7** (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of $x$ up to and including the term in $x^3$

Where appropriate express each coefficient as an exact fraction in its lowest terms.

**(3)**

(b) Write down the range of values of $x$ for which your expression is valid.

**(1)**

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where $P$ and $Q$ are rational numbers whose values should be stated.

**(2)**

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of $x$ up to and including the term in $x^2$

**(2)**

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

**(3)**

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## 16. Question 7 · part a · 3 marks · Working and answer

Working and answer

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## 17. Question 5 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

**5** (a) Expand $\sqrt{1-x}$ in ascending powers of $x$ up to and including the term in $x^3$
Give each coefficient as an exact fraction in its lowest terms.

**(3)**

(b) Using your expansion with a suitable value of $x$, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

**(3)**

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

**(2)**

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## 17. Question 5 · part a · 3 marks · Working and answer

Working and answer

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## 18. Question 9 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

**9** (a) Expand $\left(1-8x^2\right)^{\frac{1}{2}}$ in ascending powers of $x$, up to and including the term in $x^6$ giving each coefficient as an integer.

(3)

$$g(x)=\frac{a+bx}{\sqrt{1-8x^2}} \quad \text{where } a \text{ and } b \text{ are prime numbers}$$

Given that the fourth and fifth terms, in ascending powers of $x$, in the series expansion of $g(x)$ are $20x^3$ and $48x^4$ respectively,

(b) find the value of $a$ and the value of $b$

(4)

Using the first five terms, in ascending powers of $x$, in the series expansion of $g(x)$

(c) obtain an estimate, to 4 significant figures, of $\int_0^{0.2} g(x)\,dx$

(4)

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## 18. Question 9 · part a · 3 marks · Working and answer

Working and answer

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# Practice workbook · When is a binomial expansion valid?

Stating the range of x for which a binomial series with a rational power converges.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-19

# 1. Question 7 · part b · 1 mark

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.

**7** (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of $x$ up to and including the term in $x^3$

Where appropriate express each coefficient as an exact fraction in its lowest terms.

**(3)**

(b) Write down the range of values of $x$ for which your expression is valid.

**(1)**

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where $P$ and $Q$ are rational numbers whose values should be stated.

**(2)**

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of $x$ up to and including the term in $x^2$

**(2)**

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

**(3)**

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## 1. Question 7 · part b · 1 mark · Working and answer

Working and answer

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## 2. Question 9 · part b · 1 mark

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

**9** (a) Expand $(1 + 2x)^{-\frac{1}{3}}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as a fraction in its lowest terms.

**(3)**

(b) Find the range of values of $x$ for which your expansion is valid.

**(1)**

$$f(x) = \frac{2 + kx^2}{(1 + 2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f $(x)$ in ascending powers of $x$ up to and including the term in $x^3$
Give your coefficients in terms of $k$ where appropriate.

**(3)**

The coefficient of $x^3$ in the series expansion of f $(x)$ is $-\frac{8}{3}$

(d) Find the exact value of $k$

**(2)**

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\, dx$$

Give your answer to 4 decimal places.

**(5)**

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## 2. Question 9 · part b · 1 mark · Working and answer

Working and answer

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## 3. Question 2 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**2** (a) Expand $\frac{2}{\sqrt{1+3x}}$ in ascending powers of $x$ up to and including the term in $x^3$

Express each coefficient as a fraction in its simplest terms where appropriate.

**(4)**

(b) State the range of values of $x$ for which the expansion is valid.

**(1)**

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## 3. Question 2 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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**Question 2 continued**

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**(Total for Question 2 is 5 marks)**

## 3. Question 2 · part b · 1 mark · Working and answer

Working and answer

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# Practice workbook · Getting a bracket ready for the binomial series: a(1 + bx)^n

Rewriting a bracket such as (4 + 3x)^(1/2) or (2 - x)^(-3) as a constant times (1 + bx)^n, ready for a binomial expansion.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-02

# 1. Question 5 · part a · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a only.

**5** Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where $p$ and $q$ are constants,

(a) find the value of $p$ and the value of $q$

**(2)**

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of $x$ up to and including the term in $x^3$, expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of $x$ up to and including the term in $x^3$, expressing each coefficient as an exact fraction in its lowest terms.

**(2)**

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

**(4)**

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## 1. Question 5 · part a · 2 marks · Working and answer

Working and answer

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## 2. Question 6 · part a · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

**6** Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where $p$ and $q$ are constants,

(a) find the value of $p$ and the value of $q$

**(2)**

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of $x$ up to and including the term in $x^2$, expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

Using the expansion found in part (b) with a suitable value of $x$

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

**(2)**

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## 2. Question 6 · part a · 2 marks · Working and answer

Working and answer

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## 3. Question 6 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

**6** (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where $A$ and $B$ are integers to be found.

**(2)**

(b) Hence expand $\sqrt{4-x}$ in ascending powers of $x$ up to and including the term in $x^3$
Give each coefficient as an exact fraction in its lowest terms.

**(3)**

(c) Use your expansion with a suitable value of $x$ to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

**(3)**

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## 3. Question 6 · part a · 2 marks · Working and answer

Working and answer

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## 4. Question 9 · part a · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

**9** Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of $p$ and the value of $q$.

**(2)**

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of $x$ up to and including the term in $x^3$ and express each coefficient as an exact fraction in its lowest terms.

**(3)**

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of $a$ and the value of $b$.

**(3)**

(d) Find the exact value of $c$.

**(2)**

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## 4. Question 9 · part a · 2 marks · Working and answer

Working and answer

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## 5. Question 9 · part a · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

**9** (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where $a$ and $b$ are fractions in their lowest terms.

**(2)**

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of $x$ up to and including the term in $x^3$
Express each coefficient as a fraction in its lowest terms.

**(3)**

(c) (i) Use a suitable value of $x$ with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

**(4)**

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## 5. Question 9 · part a · 2 marks · Working and answer

Working and answer

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## 6. Question 6 · part a · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

**6** (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where $a$ and $b$ are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of $x$ can be written as

$$P+Qx+Rx^2+Sx^3$$

where $P$, $Q$, $R$ and $S$ are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find $P$ and $R$ in terms of $a$ and $b$, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of $x$, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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## 6. Question 6 · part a · 2 marks · Working and answer

Working and answer

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## 7. Question 6 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

**6** Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where $p$ and $q$ are constants

(a) find the value of $p$ and the value of $q$.

**(2)**

(b) Hence expand $\sqrt{9-x}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

Using the expansion you found in part (b) with a suitable value of $x$,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

**(3)**

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## 7. Question 6 · part a · 2 marks · Working and answer

Working and answer

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## 8. Question 10 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

**10** (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-2}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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## 8. Question 10 · part a · 2 marks · Working and answer

Working and answer

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## 9. Question 7 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

**7** (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of $x$ up to and including the term in $x^3$

Where appropriate express each coefficient as an exact fraction in its lowest terms.

**(3)**

(b) Write down the range of values of $x$ for which your expression is valid.

**(1)**

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where $P$ and $Q$ are rational numbers whose values should be stated.

**(2)**

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of $x$ up to and including the term in $x^2$

**(2)**

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

**(3)**

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## 9. Question 7 · part c · 2 marks · Working and answer

Working and answer

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Practice workbook · Finding the first term and common difference of an arithmetic series

Using the nth term a + (n - 1)d to set up and solve equations for a and d, or to find a particular term or unknown.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-01

# 1. Question 2 · part a · 5 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part a only.

**2** The sum of the fifth, sixth and seventh terms of an arithmetic series $A$ is nine times the sum of the first and second terms.
The third term of $A$ is 12

(a) Find the first term and common difference of $A$

**(5)**

The $n$th term of $A$ is $u_n$

(b) Find the value of $\sum_{r=15}^{60} u_r$

**(4)**

The sum to $n$ terms of $A$ is $S_n$

Given that $2S_n - 5u_n = 10$

(c) find the value of $n$

**(4)**

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## 1. Question 2 · part a · 5 marks · Working and answer

Working and answer

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## 2. Question 6 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

**6** An arithmetic series $A$ has first term $a$ and common difference $d$.

The sum $S_n$ of the first $n$ terms of $A$ is given by $S_n = n(15 + 2n)$

(a) Find the value of $a$ and the value of $d$.

**(4)**

(b) Find the 20th term of $A$.

**(2)**

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of $p$.

**(4)**

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## 2. Question 6 · part b · 2 marks · Working and answer

Working and answer

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## 3. Question 1 · part whole · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part whole only.

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**1** The $n$th term of an arithmetic series is $a_n$ where

$$a_{10} + a_{11} + a_{12} = 129 \quad \text{and} \quad a_{19} + a_{20} + a_{21} = 237$$

Find $a_1$

**(4)**

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**(Total for Question 1 is 4 marks)**

## 3. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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## 4. Question 7 · part a.i · 0 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

**7** An arithmetic series $P$ has first term $a$, common difference $d$ and $n$th term $u_n$

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of $a$

**(4)**

Given further that $u_9 = 42$

(b) find the value of $x$

**(2)**

The sum of the first $n$ terms of $P$ is $S_n$

(c) Find the value of $n$ for which $S_{(n+1)} = 12u_n + 18$

**(5)**

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## 4. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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## 5. Question 7 · part a.ii · 4 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

**7** An arithmetic series $P$ has first term $a$, common difference $d$ and $n$th term $u_n$

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of $a$

**(4)**

Given further that $u_9 = 42$

(b) find the value of $x$

**(2)**

The sum of the first $n$ terms of $P$ is $S_n$

(c) Find the value of $n$ for which $S_{(n+1)} = 12u_n + 18$

**(5)**

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## 5. Question 7 · part a.ii · 4 marks · Working and answer

Working and answer

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## 6. Question 7 · part b · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

**7** An arithmetic series $P$ has first term $a$, common difference $d$ and $n$th term $u_n$

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of $a$

**(4)**

Given further that $u_9 = 42$

(b) find the value of $x$

**(2)**

The sum of the first $n$ terms of $P$ is $S_n$

(c) Find the value of $n$ for which $S_{(n+1)} = 12u_n + 18$

**(5)**

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## 6. Question 7 · part b · 2 marks · Working and answer

Working and answer

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## 7. Question 5 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

**5** The sum of the first 10 terms of an arithmetic series $A$ is $36k + 1$ where $k$ is a constant.

The 6th term of $A$ is $4k + 1$

(a) (i) Find an expression in terms of $k$ for the common difference of $A$

(ii) Show that the first term of $A$ is $-8$

**(5)**

Given that the 4th term of $A$ is 7

(b) show that $k = 4$

**(2)**

The sum of the first $n$ terms of $A$ is $S_n$ and the $n$th term of $A$ is $U_n$

(c) Find the value of $n$ such that $S_n = 5U_{n+10} + 105$

**(4)**

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## 7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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## 8. Question 1 · part a.i · 0 marks

2020 JAN · Paper 01

Further practice · Attempt part a.i only.

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**1** The $n$th term of an arithmetic series is $t_n$ and the common difference of the series is $d$.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

**(4)**

A different arithmetic series $A$ has first term 24 and common difference 6
For series $A$, the sum of the first $2n$ terms is 3 times the sum of the first $n$ terms.

(b) Find the value of $n$.

**(5)**

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## 8. Question 1 · part a.i · 0 marks · Working and answer

Working and answer

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## 9. Question 1 · part a.ii · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part a.ii only.

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**1** The $n$th term of an arithmetic series is $t_n$ and the common difference of the series is $d$.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

**(4)**

A different arithmetic series $A$ has first term 24 and common difference 6
For series $A$, the sum of the first $2n$ terms is 3 times the sum of the first $n$ terms.

(b) Find the value of $n$.

**(5)**

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## 9. Question 1 · part a.ii · 4 marks · Working and answer

Working and answer

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## 10. Question 2 · part a.i · 0 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

**2** The $n$th term of an arithmetic series is $u_n$

Given that $u_5 = 46$ and that $u_{20} = 181$

(a) find

(i) the common difference of this series,

(ii) the first term of this series.

**(4)**

(b) Evaluate $\sum_{n=21}^{50} u_n$

**(4)**

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## 10. Question 2 · part a.i · 0 marks · Working and answer

Working and answer

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## 11. Question 2 · part a.ii · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

**2** The $n$th term of an arithmetic series is $u_n$

Given that $u_5 = 46$ and that $u_{20} = 181$

(a) find

(i) the common difference of this series,

(ii) the first term of this series.

**(4)**

(b) Evaluate $\sum_{n=21}^{50} u_n$

**(4)**

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## 11. Question 2 · part a.ii · 4 marks · Working and answer

Working and answer

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## 12. Question 8 · part a · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

**8** The sum to $n$ terms of an arithmetic series $A$ is $S_n$

The sum of the first four terms of $A$ is 42 and the fifth term of $A$ is 23

(a) Show that $S_n = \sum_{r=1}^{n}(Pr - Q)$ where $P$ and $Q$ are prime numbers.

**(6)**

$S_{2n} - 3U_n = 1062$ where $U_n$ is the $n$th term of $A$

(b) Find the value of $n$

**(4)**

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## 12. Question 8 · part a · 6 marks · Working and answer

Working and answer

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## 13. Question 1 · part whole · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

**1** An arithmetic series has 5th term 16 and 100th term 301

Find the sum of the first 50 terms of the series.

**(5)**

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**(Total for Question 1 is 5 marks)**

## 13. Question 1 · part whole · 5 marks · Working and answer

Working and answer

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## 14. Question 5 · part a.i · 0 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

**5** The sum of the first 10 terms of an arithmetic series $A$ is $36k + 1$ where $k$ is a constant.

The 6th term of $A$ is $4k + 1$

(a) (i) Find an expression in terms of $k$ for the common difference of $A$

(ii) Show that the first term of $A$ is $-8$

**(5)**

Given that the 4th term of $A$ is 7

(b) show that $k = 4$

**(2)**

The sum of the first $n$ terms of $A$ is $S_n$ and the $n$th term of $A$ is $U_n$

(c) Find the value of $n$ such that $S_n = 5U_{n+10} + 105$

**(4)**

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## 14. Question 5 · part a.i · 0 marks · Working and answer

Working and answer

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## 15. Question 5 · part a.ii · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

**5** The sum of the first 10 terms of an arithmetic series $A$ is $36k + 1$ where $k$ is a constant.

The 6th term of $A$ is $4k + 1$

(a) (i) Find an expression in terms of $k$ for the common difference of $A$

(ii) Show that the first term of $A$ is $-8$

**(5)**

Given that the 4th term of $A$ is 7

(b) show that $k = 4$

**(2)**

The sum of the first $n$ terms of $A$ is $S_n$ and the $n$th term of $A$ is $U_n$

(c) Find the value of $n$ such that $S_n = 5U_{n+10} + 105$

**(4)**

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## 15. Question 5 · part a.ii · 5 marks · Working and answer

Working and answer

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# Practice workbook · Using a binomial expansion to estimate a value

Choosing a value of x that turns the bracket into the required number and substituting into the truncated series to estimate it.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-20

## 1. Question 9 · part b · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

**9** (a) Expand $\sqrt{1-4x}$ in ascending powers of $x$ up to and including the term in $x^3$, giving each coefficient as an integer.

**(3)**

(b) Use your expansion with a suitable value for $x$ to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

**(3)**

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

**(2)**

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## 1. Question 9 · part b · 3 marks · Working and answer

Working and answer

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## 2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

**9** (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of $x$ up to and including the term in $x^3$

Simplify each term as far as possible.

**(3)**

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

**(2)**

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where $a$, $b$ and $c$ are integers.

Show all your working clearly.

**(2)**

Using the expansion you found in part (a) with a suitable value of $x$

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

**(4)**

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## 2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

**Question 9 continued**

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## 2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

**Question 9 continued**

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## 2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

**Question 9 continued**

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**(Total for Question 9 is 11 marks)**

## 2. Question 9 · part d · 4 marks · Working and answer

Working and answer

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## 3. Question 6 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

**6** Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where $p$ and $q$ are constants,

(a) find the value of $p$ and the value of $q$

**(2)**

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of $x$ up to and including the term in $x^2$, expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

Using the expansion found in part (b) with a suitable value of $x$

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

**(2)**

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## 3. Question 6 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

**Question 6 continued**

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**(Total for Question 6 is 7 marks)**

## 3. Question 6 · part c · 2 marks · Working and answer

Working and answer

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## 4. Question 6 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

**6** (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where $A$ and $B$ are integers to be found.

**(2)**

(b) Hence expand $\sqrt{4-x}$ in ascending powers of $x$ up to and including the term in $x^3$
Give each coefficient as an exact fraction in its lowest terms.

**(3)**

(c) Use your expansion with a suitable value of $x$ to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

**(3)**

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## 4. Question 6 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

**Question 6 continued**

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**(Total for Question 6 is 8 marks)**

## 4. Question 6 · part c · 3 marks · Working and answer

Working and answer

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## 5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

**6** Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where $p$ and $q$ are constants

(a) find the value of $p$ and the value of $q$.

**(2)**

(b) Hence expand $\sqrt{9-x}$ in ascending powers of $x$ up to and including the term in $x^3$ expressing each coefficient as an exact fraction in its lowest terms.

**(3)**

Using the expansion you found in part (b) with a suitable value of $x$,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

**(3)**

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## 5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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**Question 6 continued**

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## 5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

**Question 6 continued**

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## 5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

**Question 6 continued**

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**(Total for Question 6 is 8 marks)**

## 5. Question 6 · part c · 3 marks · Working and answer

Working and answer

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## 6. Question 1 · part b · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**1** (a) Expand $\left(1+\frac{x}{4}\right)^8$ in ascending powers of $x$ up to and including the term in $x^3$

Give each coefficient in its simplest terms.

**(3)**

(b) Use your expansion with a suitable value of $x$ to obtain an approximation, to 4 decimal places, of $(1.035)^8$

**(3)**

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**(Total for Question 1 is 6 marks)**

## 6. Question 1 · part b · 3 marks · Working and answer

Working and answer

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## 7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

**9** (a) Expand $\sqrt{1-4x}$ in ascending powers of $x$ up to and including the term in $x^3$, giving each coefficient as an integer.

**(3)**

(b) Use your expansion with a suitable value for $x$ to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

**(3)**

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

**(2)**

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## 7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

**Question 9 continued**

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## 7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

**Question 9 continued**

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## 7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.



**Question 9 continued**

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**(Total for Question 9 is 8 marks)**

## 7. Question 9 · part c · 2 marks · Working and answer

Working and answer

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## 8. Question 9 · part c.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.i only.

**9** (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where $a$ and $b$ are fractions in their lowest terms.

**(2)**

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of $x$ up to and including the term in $x^3$
Express each coefficient as a fraction in its lowest terms.

**(3)**

(c) (i) Use a suitable value of $x$ with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

**(4)**

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## 8. Question 9 · part c.i · 0 marks · Working and answer

Working and answer

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# 9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

**9** (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where $a$ and $b$ are fractions in their lowest terms.

**(2)**

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of $x$ up to and including the term in $x^3$
Express each coefficient as a fraction in its lowest terms.

**(3)**

(c) (i) Use a suitable value of $x$ with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

**(4)**

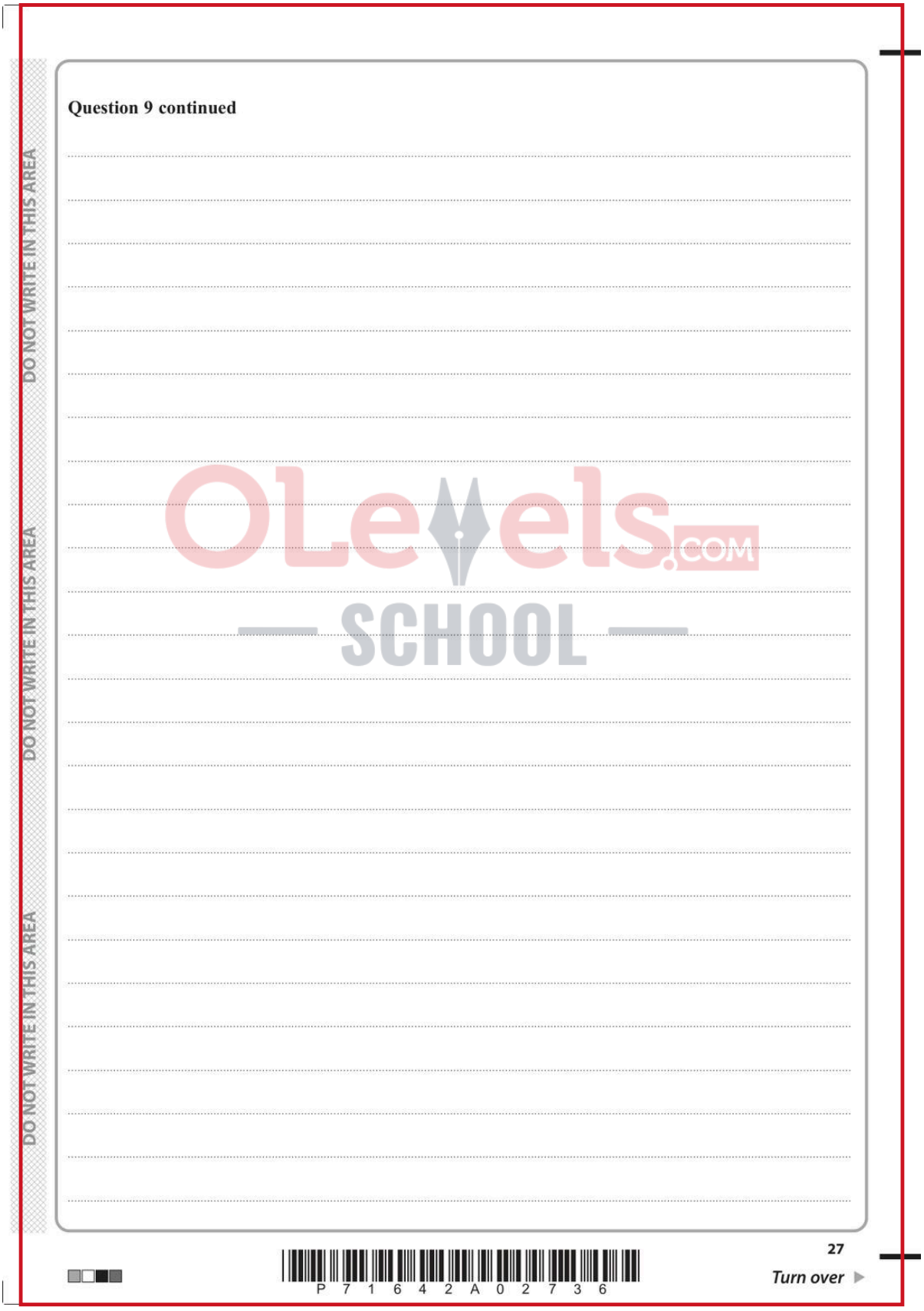
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## 9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.



## 9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

**Question 9 continued**

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## 9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.



## 9. Question 9 · part c.ii · 4 marks · Working and answer

Working and answer

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## 10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

**6** (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where $a$ and $b$ are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of $x$ can be written as

$$P+Qx+Rx^2+Sx^3$$

where $P$, $Q$, $R$ and $S$ are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find $P$ and $R$ in terms of $a$ and $b$, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of $x$, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

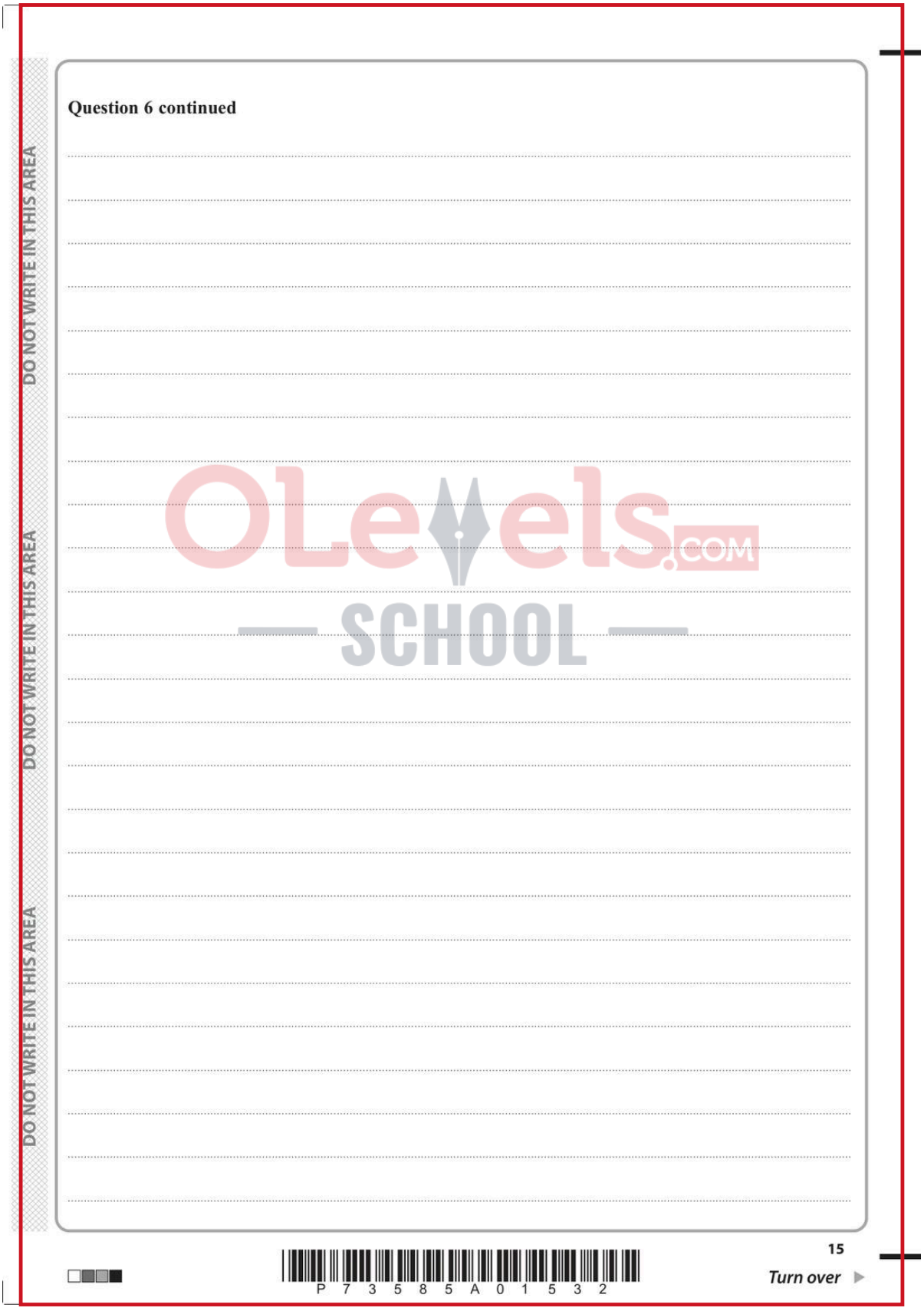
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## 10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.



**Question 6 continued**

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Turn over

## 10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

**Question 6 continued**

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## 10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

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**Question 6 continued**

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**(Total for Question 6 is 12 marks)**

## 10. Question 6 · part d · 3 marks · Working and answer

Working and answer

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## 11. Question 5 · part b · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

**5** (a) Expand $\sqrt{1-x}$ in ascending powers of $x$ up to and including the term in $x^3$
Give each coefficient as an exact fraction in its lowest terms.

**(3)**

(b) Using your expansion with a suitable value of $x$, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

**(3)**

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

**(2)**

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## 11. Question 5 · part b · 3 marks · Working and answer

Working and answer

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## 12. Question 5 · part c · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

**5** (a) Expand $\sqrt{1-x}$ in ascending powers of $x$ up to and including the term in $x^3$
Give each coefficient as an exact fraction in its lowest terms.

**(3)**

(b) Using your expansion with a suitable value of $x$, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

**(3)**

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

**(2)**

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## 12. Question 5 · part c · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.



**Question 5 continued**

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**(Total for Question 5 is 8 marks)**

## 12. Question 5 · part c · 2 marks · Working and answer

Working and answer

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# Practice workbook · Finding n from an equation linking sums and terms

Substituting the sum and nth-term formulas into a given equation and solving the quadratic for a positive whole number n.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-06

# 1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

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**1** The $n$th term of an arithmetic series is $t_n$ and the common difference of the series is $d$.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

**(4)**

A different arithmetic series $A$ has first term 24 and common difference 6
For series $A$, the sum of the first $2n$ terms is 3 times the sum of the first $n$ terms.

(b) Find the value of $n$.

**(5)**

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# 1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

**Question 1 continued**

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## 1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

**Question 1 continued**

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**(Total for Question 1 is 9 marks)**

## 1. Question 1 · part b · 5 marks · Working and answer

Working and answer

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## 2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**6** (a) Show that $\sum_{r=1}^{n}(4r-3) = n(2n-1)$

**(3)**

(b) Hence, or otherwise, find the least value of $n$ such that $\sum_{r=1}^{n}(4r-3) > 1000$

**(3)**

Given that $S_n = n(2n-1)$, $t_n = (4n-3)$ and that $18 + 3t_{n+7} = S_{n+4}$

(c) find the value of $n$.

**(4)**

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## 2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 6 continued**

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## 2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 6 continued**

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## 2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 6 continued**

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**(Total for Question 6 is 10 marks)**

## 2. Question 6 · part c · 4 marks · Working and answer

Working and answer

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## 3. Question 6 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

**6** An arithmetic series $A$ has first term $a$ and common difference $d$.

The sum $S_n$ of the first $n$ terms of $A$ is given by $S_n = n(15 + 2n)$

(a) Find the value of $a$ and the value of $d$.

**(4)**

(b) Find the 20th term of $A$.

**(2)**

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of $p$.

**(4)**

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## 3. Question 6 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.



## 3. Question 6 · part c · 4 marks · Working and answer

Working and answer

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## 4. Question 8 · part b · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

**8** The sum to $n$ terms of an arithmetic series $A$ is $S_n$

The sum of the first four terms of $A$ is 42 and the fifth term of $A$ is 23

(a) Show that $S_n = \sum_{r=1}^{n}(Pr - Q)$ where $P$ and $Q$ are prime numbers.

**(6)**

$S_{2n} - 3U_n = 1062$ where $U_n$ is the $n$th term of $A$

(b) Find the value of $n$

**(4)**

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## 4. Question 8 · part b · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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**Question 8 continued**

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**(Total for Question 8 is 10 marks)**

## 4. Question 8 · part b · 4 marks · Working and answer

Working and answer

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## 5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

**6** The sum of the first $n$ terms of an arithmetic series is $S_n$ where $S_n = n(3 + 2n)$

(a) Find the value of $S_{20}$

**(2)**

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of $A$ and the value of $B$

**(6)**

A different arithmetic series has first term 7 and common difference 4

The sum of the first $n$ terms of this series is $T_n$

(c) Use algebra to find the value of $n$ for which $T_n = S_n + 252$

**(5)**

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## 5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

**Question 6 continued**

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## 5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

**Question 6 continued**

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## 5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

**Question 6 continued**

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**(Total for Question 6 is 13 marks)**

## 5. Question 6 · part c · 5 marks · Working and answer

Working and answer

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## 6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

**7** An arithmetic series $P$ has first term $a$, common difference $d$ and $n$th term $u_n$

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of $a$

**(4)**

Given further that $u_9 = 42$

(b) find the value of $x$

**(2)**

The sum of the first $n$ terms of $P$ is $S_n$

(c) Find the value of $n$ for which $S_{(n+1)} = 12u_n + 18$

**(5)**

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## 6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 7 continued**

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## 6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 7 continued**

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## 6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.



## 6. Question 7 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

**9** (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first $n$ terms of an arithmetic series is $S_n$ where $S_n=\sum_{r=1}^{n}(5r-1)$

The $r$th term of this series is $u_r$

Given that $S_n=12u_{n+1}+52$

(c) find the value of $n$ (5)

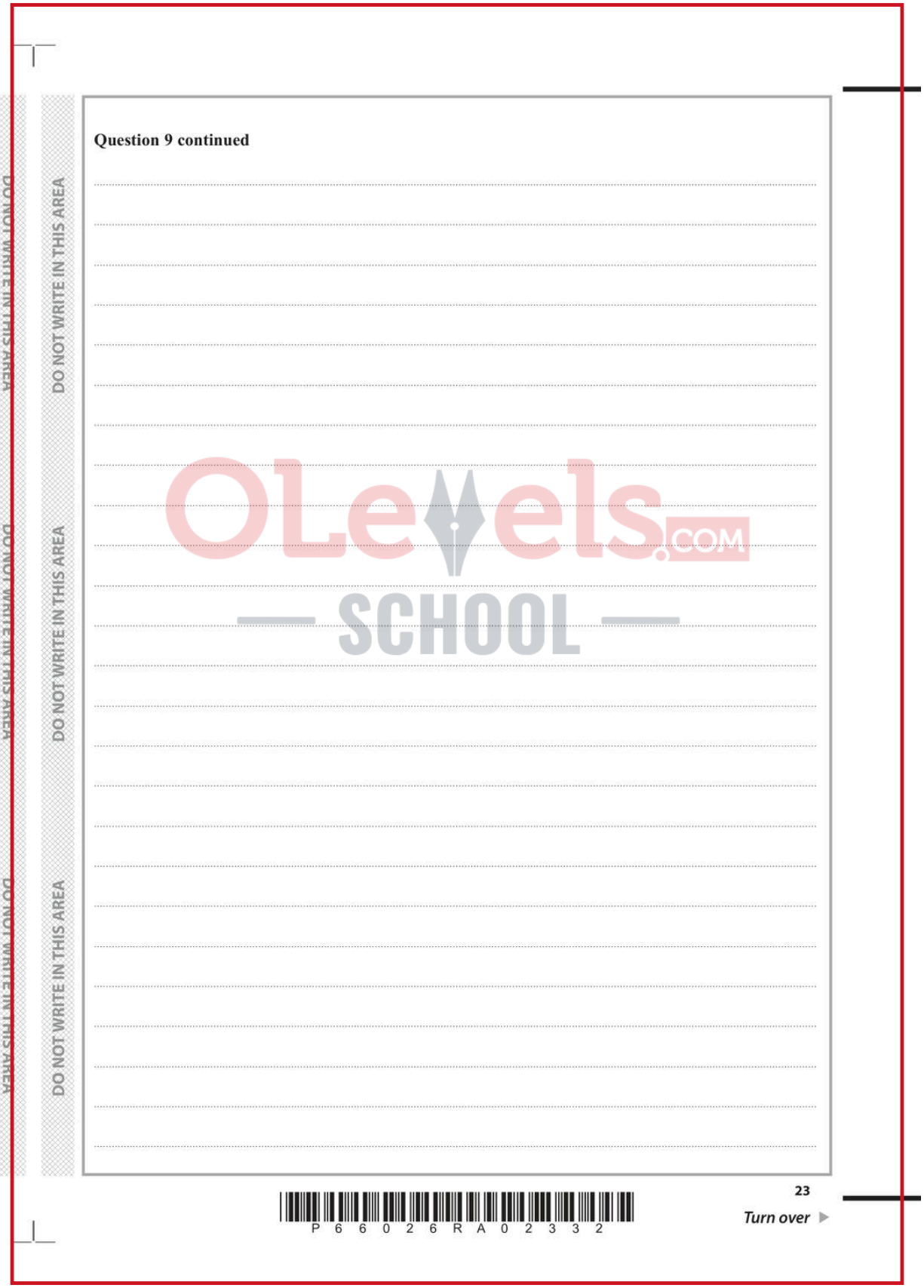
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## 7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.



## 7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

**Question 9 continued**

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## 7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

**Question 9 continued**

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**(Total for Question 9 is 11 marks)**

## 7. Question 9 · part c · 5 marks · Working and answer

Working and answer

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## 8. Question 2 · part c · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

**2** The sum of the fifth, sixth and seventh terms of an arithmetic series $A$ is nine times the sum of the first and second terms.
The third term of $A$ is 12

(a) Find the first term and common difference of $A$

**(5)**

The $n$th term of $A$ is $u_n$

(b) Find the value of $\sum_{r=15}^{60} u_r$

**(4)**

The sum to $n$ terms of $A$ is $S_n$

Given that $2S_n - 5u_n = 10$

(c) find the value of $n$

**(4)**

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## 8. Question 2 · part c · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

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**Question 2 continued**

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**(Total for Question 2 is 13 marks)**

## 8. Question 2 · part c · 4 marks · Working and answer

Working and answer

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## 9. Question 5 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

**5** (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of $n$. (3)

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## 9. Question 5 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

**Question 5 continued**

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**(Total for Question 5 is 8 marks)**

## 9. Question 5 · part c · 3 marks · Working and answer

Working and answer

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## 10. Question 3 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

**3** (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of $n$ (3)

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## 10. Question 3 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

**Question 3 continued**

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**(Total for Question 3 is 8 marks)**

## 10. Question 3 · part c · 3 marks · Working and answer

Working and answer

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## 11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**5** The sum of the first 10 terms of an arithmetic series $A$ is $36k + 1$ where $k$ is a constant.

The 6th term of $A$ is $4k + 1$

(a) (i) Find an expression in terms of $k$ for the common difference of $A$

(ii) Show that the first term of $A$ is $-8$

**(5)**

Given that the 4th term of $A$ is 7

(b) show that $k = 4$

**(2)**

The sum of the first $n$ terms of $A$ is $S_n$ and the $n$th term of $A$ is $U_n$

(c) Find the value of $n$ such that $S_n = 5U_{n+10} + 105$

**(4)**

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## 11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 5 continued**

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## 11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 5 continued**

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## 11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 5 continued**

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**(Total for Question 5 is 11 marks)**

## 11. Question 5 · part c · 4 marks · Working and answer

Working and answer

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# Practice workbook · Finding the first term and common ratio of a geometric series

Writing given terms as ar^(n-1), dividing to remove a, and solving for r and a, or finding a term from the nth-term formula.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-08

# 1. Question 4 · part a.ii · 5 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

**4** A geometric series $G$ has first term $a$ and common ratio $r$

The third term of $G$ is 10 and the seventh term of $G$ is 40

Given that the second term of $G$ is negative,

(a) find

(i) the exact value of $r$

(ii) the value of $a$

**(5)**

(b) Explain why $G$ does not have a sum to infinity.

**(1)**

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## 1. Question 4 · part a.ii · 5 marks · Working and answer

Working and answer

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## 2. Question 3 · part a.i · 0 marks

2022 JAN · Paper 01

Further practice · Attempt part a.i only.

**3** A geometric series $G$ has first term $a$ and common ratio $r$

The 2nd term of $G$ is $\frac{5}{16}$ and the 5th term of $G$ is $\frac{135}{1024}$

(a) Find

(i) the value of $r$

(ii) the value of $a$

**(5)**

Given that $G$ is convergent with sum to infinity $S$

(b) find the exact value of $S$

**(2)**

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## 2. Question 3 · part a.i · 0 marks · Working and answer

Working and answer

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## 3. Question 3 · part a.ii · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part a.ii only.

**3** A geometric series $G$ has first term $a$ and common ratio $r$

The 2nd term of $G$ is $\frac{5}{16}$ and the 5th term of $G$ is $\frac{135}{1024}$

(a) Find

(i) the value of $r$

(ii) the value of $a$

**(5)**

Given that $G$ is convergent with sum to infinity $S$

(b) find the exact value of $S$

**(2)**

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## 3. Question 3 · part a.ii · 5 marks · Working and answer

Working and answer

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## 4. Question 7 · part a.i · 0 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

**7** The $n$th term of a geometric series $G$ is $u_n$

The first term of $G$ is $a$ and the common ratio of $G$ is $r$, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of $a$.

**(3)**

(b) Find the least value of $n$ for which $u_n > 500$

**(4)**

The sum of the first $n$ terms of $G$ is $S_n$

(c) Find $S_{20}$

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where $p$ is an integer.

**(4)**

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## 4. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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## 5. Question 7 · part a.ii · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

**7** The $n$th term of a geometric series $G$ is $u_n$

The first term of $G$ is $a$ and the common ratio of $G$ is $r$, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of $a$.

**(3)**

(b) Find the least value of $n$ for which $u_n > 500$

**(4)**

The sum of the first $n$ terms of $G$ is $S_n$

(c) Find $S_{20}$

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where $p$ is an integer.

**(4)**

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## 5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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## 6. Question 5 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**5** A geometric series $G$ has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of $G$

**(2)**

(b) Show that the 6th term of $G$ can be written as $\frac{3^6}{2^{13}}$

**(3)**

The $n$th term of $G$ is $u_n$

(c) By finding an expression for $u_n$ in terms of $n$, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

**(5)**

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## 6. Question 5 · part b · 3 marks · Working and answer

Working and answer

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## 7. Question 4 · part a.i · 0 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

**4** A geometric series $G$ has first term $a$ and common ratio $r$

The third term of $G$ is 10 and the seventh term of $G$ is 40

Given that the second term of $G$ is negative,

(a) find

(i) the exact value of $r$

(ii) the value of $a$

**(5)**

(b) Explain why $G$ does not have a sum to infinity.

**(1)**

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## 7. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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## 8. Question 7 · part a · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

**7** The geometric series $G$ has first term $a$, common ratio $r$ and $n$th term $u_n$

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

**(3)**

(b) Hence find $a$ in terms of e and $x$.

**(3)**

Given that the sum to infinity of $G$ can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for $p$ in terms of $x$.

**(3)**

Given that $u_{18} > 1.6$ and that $x$ is an integer,

(d) find the least value of $x$.

**(4)**

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## 8. Question 7 · part a · 3 marks · Working and answer

Working and answer

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## 9. Question 7 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**7** The geometric series $G$ has first term $a$, common ratio $r$ and $n$th term $u_n$

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

**(3)**

(b) Hence find $a$ in terms of e and $x$.

**(3)**

Given that the sum to infinity of $G$ can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for $p$ in terms of $x$.

**(3)**

Given that $u_{18} > 1.6$ and that $x$ is an integer,

(d) find the least value of $x$.

**(4)**

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**20**

## 9. Question 7 · part b · 3 marks · Working and answer

Working and answer

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## 10. Question 3 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

**3** A geometric series has first term $a$ and common ratio $r$, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of $r$

(ii) the value of $a$

**(4)**

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where $p$ and $q$ are integers.

**(2)**

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## 10. Question 3 · part a.i · 0 marks · Working and answer

Working and answer

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## 11. Question 3 · part a.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

**3** A geometric series has first term $a$ and common ratio $r$, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of $r$

(ii) the value of $a$

**(4)**

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where $p$ and $q$ are integers.

**(2)**

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## 11. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

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## 12. Question 7 · part a · 3 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

**7** The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

**(3)**

A different geometric series $G$ has a common ratio $r$ and $n$th term $t_n$

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that $r$ satisfies the equation

$$2r^2 + 5r + 2 = 0$$

**(5)**

Given further that $G$ is convergent with sum to infinity $S$,

(c) find the value of $S$.

**(4)**

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## 12. Question 7 · part a · 3 marks · Working and answer

Working and answer

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## 13. Question 8 · part a · 1 mark

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

**8** The $n$th term of a geometric series $G$ is $U_n$ and the sum of the first $n$ terms of $G$ is $S_n$

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of $U_5$

**(1)**

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where $A$ and $B$ are integers to be found.

**(3)**

The sum to infinity of $G$ is $S$

(c) Find the least value of $n$ such that $S - S_n < 0.045$

**(6)**

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## 13. Question 8 · part a · 1 mark · Working and answer

Working and answer

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## 14. Question 6 · part a · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

**6** (i) A geometric series $G$ has first term $\left(3+\sqrt{3}\right)$ and common ratio $r$

The second term of $G$ is $\sqrt{3}$

(a) Find, showing all your working, the exact value of $r$

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where $p$ and $q$ are integers to be found.

**(2)**

(b) Explain why $G$ is convergent.

**(1)**

(ii) A different geometric series $H$ has first term 8 and common ratio 0.6

The sum to $n$ terms of $H$ is $S_n$ and the sum to infinity of $H$ is $T$

Find, using logarithms, the least value of $n$ such that $T - S_n < 0.12$

**(5)**

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## 14. Question 6 · part a · 2 marks · Working and answer

Working and answer

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## 15. Question 1 · part whole · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

**1** The $n$th term of an arithmetic series $A$ is $a_n$

The $n$th term of a geometric series $G$ is $t_n$

For these two series

$$a_1 = t_1 \qquad a_{10} = t_3 = 48 \qquad a_{10} = 4t_2$$

Find

(i) the common ratio of $G$,

(ii) the common difference of $A$.

**(6)**

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**(Total for Question 1 is 6 marks)**

## 15. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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## 16. Question 7 · part a · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

**7** A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of $x$.

**(5)**

Given that $x < 1$,

(b) show that the series is convergent.

**(2)**

The sum to infinity of the series is $S$.

(c) Find the value of $S$.

**(2)**

The sum of the first $n$ terms of the series is $S_n$

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of $n$.

**(3)**

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## 16. Question 7 · part a · 5 marks · Working and answer

Working and answer

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## 17. Question 6 · part a · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part a only.

**6** The $n$th term of a geometric series $G$ is $U_n$

The first three terms of $G$ are given by

$$U_1 = q(4p + 1) \qquad U_2 = q(2p + 3) \qquad U_3 = q(2p - 3)$$

(a) Find the possible values of $p$

**(5)**

Given that $G$ is convergent with sum to infinity 250

(b) find the value of $q$

**(3)**

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## 17. Question 6 · part a · 5 marks · Working and answer

Working and answer

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# Practice workbook · Does a geometric series have a sum to infinity?

Stating or showing that a geometric series has a sum to infinity because |r| < 1, or has none because |r| > 1.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-12

# 1. Question 7 · part b · 2 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

**7** A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of $x$.

**(5)**

Given that $x < 1$,

(b) show that the series is convergent.

**(2)**

The sum to infinity of the series is $S$.

(c) Find the value of $S$.

**(2)**

The sum of the first $n$ terms of the series is $S_n$

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of $n$.

**(3)**

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## 1. Question 7 · part b · 2 marks · Working and answer

Working and answer

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## 2. Question 4 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

**4** A geometric series $G$ has first term $a$ and common ratio $r$

The third term of $G$ is 10 and the seventh term of $G$ is 40

Given that the second term of $G$ is negative,

(a) find

(i) the exact value of $r$

(ii) the value of $a$

**(5)**

(b) Explain why $G$ does not have a sum to infinity.

**(1)**

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## 2. Question 4 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

**Question 4 continued**

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**(Total for Question 4 is 6 marks)**

## 2. Question 4 · part b · 1 mark · Working and answer

Working and answer

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## 3. Question 8 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**8** The sum of the first 2 terms of a geometric series $G$ is 360
The sum of the 2nd and 3rd terms of $G$ is 288

The $n$th term of $G$ is $U_n$

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where $A$ is an integer to be found.

**(7)**

(b) Explain why $G$ is convergent.

**(1)**

(c) Hence find the sum to infinity of $G$

**(2)**

(d) Find the least number of terms for which the sum is greater than 978

**(4)**

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## 3. Question 8 · part b · 1 mark · Working and answer

Working and answer

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## 4. Question 6 · part b · 1 mark

2025 NOV · Paper 02

Further practice · Attempt part b only.

**6** (i) A geometric series $G$ has first term $\left(3+\sqrt{3}\right)$ and common ratio $r$

The second term of $G$ is $\sqrt{3}$

(a) Find, showing all your working, the exact value of $r$

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where $p$ and $q$ are integers to be found.

**(2)**

(b) Explain why $G$ is convergent.

**(1)**

(ii) A different geometric series $H$ has first term 8 and common ratio 0.6

The sum to $n$ terms of $H$ is $S_n$ and the sum to infinity of $H$ is $T$

Find, using logarithms, the least value of $n$ such that $T - S_n < 0.12$

**(5)**

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## 4. Question 6 · part b · 1 mark · Working and answer

Working and answer

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# Practice workbook · Sum to infinity of a geometric series

Using S = a/(1 - r) once a and r are known or found, sometimes in surd or fraction form.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-13

# 1. Question 8 · part c · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

**8** The sum of the first and second terms of a geometric series $G$ is 400

The sum of the second and third terms of $G$ is 100

(a) Show that the common ratio of $G$ is $\frac{1}{4}$

**(4)**

(b) Show that the first term of $G$ is 320

**(2)**

(c) Find the sum to infinity of $G$

**(2)**

The sum to $n$ terms of $G$ is $S_n$

(d) Find, using logarithms, the least value of $n$ such that

$$S_n > 426.6$$

**(4)**

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## 1. Question 8 · part c · 2 marks · Working and answer

Working and answer

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## 2. Question 3 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

**3** A geometric series $G$ has first term $a$ and common ratio $r$

The 2nd term of $G$ is $\frac{5}{16}$ and the 5th term of $G$ is $\frac{135}{1024}$

(a) Find

(i) the value of $r$

(ii) the value of $a$

**(5)**

Given that $G$ is convergent with sum to infinity $S$

(b) find the exact value of $S$

**(2)**

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## 2. Question 3 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

**Question 3 continued**

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**(Total for Question 3 is 7 marks)**

## 2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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## 3. Question 5 · part a · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

**5** A geometric series $G$ has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of $G$

**(2)**

(b) Show that the 6th term of $G$ can be written as $\frac{3^6}{2^{13}}$

**(3)**

The $n$th term of $G$ is $u_n$

(c) By finding an expression for $u_n$ in terms of $n$, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

**(5)**

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## 3. Question 5 · part a · 2 marks · Working and answer

Working and answer

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## 4. Question 7 · part c · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

**7** A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of $x$.

**(5)**

Given that $x < 1$,

(b) show that the series is convergent.

**(2)**

The sum to infinity of the series is $S$.

(c) Find the value of $S$.

**(2)**

The sum of the first $n$ terms of the series is $S_n$

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of $n$.

**(3)**

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## 4. Question 7 · part c · 2 marks · Working and answer

Working and answer

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## 5. Question 6 · part b · 3 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

**6** The $n$th term of a geometric series $G$ is $U_n$

The first three terms of $G$ are given by

$$U_1 = q(4p + 1) \qquad U_2 = q(2p + 3) \qquad U_3 = q(2p - 3)$$

(a) Find the possible values of $p$

**(5)**

Given that $G$ is convergent with sum to infinity 250

(b) find the value of $q$

**(3)**

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## 5. Question 6 · part b · 3 marks

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Further practice · Attempt part b only.

**Question 6 continued**

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**(Total for Question 6 is 8 marks)**

## 5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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## 6. Question 8 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

**8** The sum of the first 2 terms of a geometric series $G$ is 360
The sum of the 2nd and 3rd terms of $G$ is 288

The $n$th term of $G$ is $U_n$

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where $A$ is an integer to be found.

**(7)**

(b) Explain why $G$ is convergent.

**(1)**

(c) Hence find the sum to infinity of $G$

**(2)**

(d) Find the least number of terms for which the sum is greater than 978

**(4)**

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## 6. Question 8 · part c · 2 marks · Working and answer

Working and answer

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## 7. Question 5 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

**5** The $n$th term of a geometric series with common ratio $r$ is $u_n$

Given that $u_2 + u_4 = 212.5$ and that $u_3 + u_4 = 62.5$

(a) find the two possible values of $r$.

**(5)**

Given that the series is convergent with sum to infinity $S$,

(b) find the exact value of $S$.

**(2)**

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## 7. Question 5 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.



## 7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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## 8. Question 7 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

**7** The geometric series $G$ has first term $a$, common ratio $r$ and $n$th term $u_n$

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

**(3)**

(b) Hence find $a$ in terms of e and $x$.

**(3)**

Given that the sum to infinity of $G$ can be written as $\dfrac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for $p$ in terms of $x$.

**(3)**

Given that $u_{18} > 1.6$ and that $x$ is an integer,

(d) find the least value of $x$.

**(4)**

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**20**

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## 8. Question 7 · part c · 3 marks · Working and answer

Working and answer

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## 9. Question 3 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**3** A geometric series has first term $a$ and common ratio $r$, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of $r$

(ii) the value of $a$

**(4)**

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where $p$ and $q$ are integers.

**(2)**

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## 9. Question 3 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**Question 3 continued**

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**(Total for Question 3 is 6 marks)**

## 9. Question 3 · part b · 2 marks · Working and answer

Working and answer

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## 10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

**7** The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

**(3)**

A different geometric series $G$ has a common ratio $r$ and $n$th term $t_n$

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that $r$ satisfies the equation

$$2r^2 + 5r + 2 = 0$$

**(5)**

Given further that $G$ is convergent with sum to infinity $S$,

(c) find the value of $S$.

**(4)**

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## 10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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## 10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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## 10. Question 7 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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**(Total for Question 7 is 12 marks)**

## 10. Question 7 · part c · 4 marks · Working and answer

Working and answer

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## 11. Question 7 · part b · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

**7** A geometric series $G$ with common ratio $r$, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of $r$

**(2)**

Given that $r > 0$

(b) find the sum to infinity of $G$

**(2)**

The sum to $n$ terms of $G$ is greater than 33

(c) Find, using logarithms, the least possible value of $n$
Show your working clearly.

**(5)**

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**20**

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## 11. Question 7 · part b · 2 marks · Working and answer

Working and answer

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## 12. Question 2 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

**2** The $n$th term of a convergent geometric series is $8^{(1-2n)}$

Find the sum to infinity of the series.

Give your answer in the form $\frac{p}{q}$ where $p$ and $q$ are integers to be found.

**(6)**

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## 12. Question 2 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

**Question 2 continued**

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**(Total for Question 2 is 6 marks)**

## 12. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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## 13. Question 4 · part ii · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

**4** (i) Find the value of $\sum_{k=6}^{17} 3(2)^{k-1}$

**(4)**

(ii) The common ratio of a geometric series $G$ is positive.

The third term of $G$ is $\frac{7}{13}$ and the ninth term of $G$ is $\frac{448}{9477}$

$G$ is convergent with sum to infinity $S$

Find the exact value of $S$

**(7)**

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## 13. Question 4 · part ii · 7 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

**Question 4 continued**

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**(Total for Question 4 is 11 marks)**

## 13. Question 4 · part ii · 7 marks · Working and answer

Working and answer

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# Practice workbook · Adding up a finite geometric series

Using S_n = a(1 - r^n)/(1 - r) (including for a sigma sum) to find or show a sum, often in exact or surd form.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-SER-11

# 1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

**7** The $n$th term of a geometric series $G$ is $u_n$

The first term of $G$ is $a$ and the common ratio of $G$ is $r$, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of $a$.

**(3)**

(b) Find the least value of $n$ for which $u_n > 500$

**(4)**

The sum of the first $n$ terms of $G$ is $S_n$

(c) Find $S_{20}$

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where $p$ is an integer.

**(4)**

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## 1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

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**Question 7 continued**

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## 1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

**Question 7 continued**

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## 1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

**Question 7 continued**

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**(Total for Question 7 is 11 marks)**

## 1. Question 7 · part c · 4 marks · Working and answer

Working and answer

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## 2. Question 4 · part i · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part i only.

**4** (i) Find the value of $\sum_{k=6}^{17} 3(2)^{k-1}$

(4)

(ii) The common ratio of a geometric series $G$ is positive.

The third term of $G$ is $\frac{7}{13}$ and the ninth term of $G$ is $\frac{448}{9477}$

$G$ is convergent with sum to infinity $S$

Find the exact value of $S$

(7)

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## 2. Question 4 · part i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 3. Question 8 · part b · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

**8** The $n$th term of a geometric series $G$ is $U_n$ and the sum of the first $n$ terms of $G$ is $S_n$

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of $U_5$

**(1)**

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where $A$ and $B$ are integers to be found.

**(3)**

The sum to infinity of $G$ is $S$

(c) Find the least value of $n$ such that $S - S_n < 0.045$

**(6)**

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## 3. Question 8 · part b · 3 marks · Working and answer

Working and answer

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# Practice workbook · Finding n in a geometric series with logarithms

Setting up an equation or inequality for a term or sum of a geometric series, isolating r^n and taking logarithms to find n.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-14

# 1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

**8** The sum of the first and second terms of a geometric series $G$ is 400

The sum of the second and third terms of $G$ is 100

(a) Show that the common ratio of $G$ is $\frac{1}{4}$

**(4)**

(b) Show that the first term of $G$ is 320

**(2)**

(c) Find the sum to infinity of $G$

**(2)**

The sum to $n$ terms of $G$ is $S_n$

(d) Find, using logarithms, the least value of $n$ such that

$$S_n > 426.6$$

**(4)**

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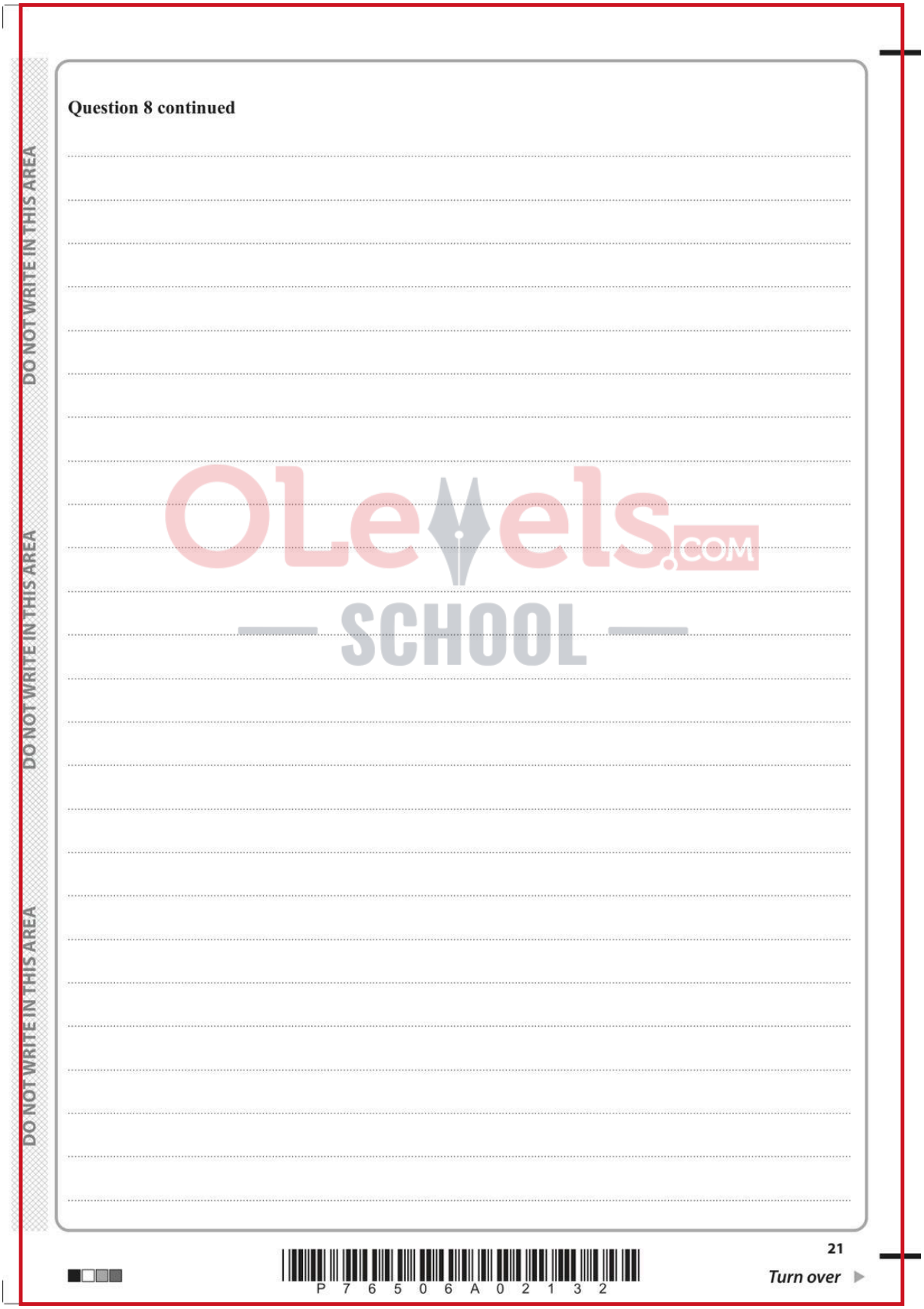
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## 1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.



# 1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

**Question 8 continued**

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## 1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

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**Question 8 continued**

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**(Total for Question 8 is 12 marks)**

## 1. Question 8 · part d · 4 marks · Working and answer

Working and answer

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## 2. Question 7 · part b · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**7** The $n$th term of a geometric series $G$ is $u_n$

The first term of $G$ is $a$ and the common ratio of $G$ is $r$, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of $a$.

**(3)**

(b) Find the least value of $n$ for which $u_n > 500$

**(4)**

The sum of the first $n$ terms of $G$ is $S_n$

(c) Find $S_{20}$

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where $p$ is an integer.

**(4)**

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## 2. Question 7 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 3. Question 9 · part b · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**9** A geometric series $G$ has common ratio $r$ where $r > 0$

The third term of $G$ is $\frac{27}{2}$ and the sum of the first three terms of $G$ is $\frac{57}{2}$

Given that the sum to $n$ terms of $G$ is $S_n$

(a) show that $S_n = \sum_{j=1}^{n} 4\left(\frac{3}{2}\right)^j$

**(8)**

Given that $S_k > 50\,000$

(b) show that the least value of $k$ is given by

$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

**(3)**

(c) Hence find the least value of $k$

**(1)**

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## 3. Question 9 · part b · 3 marks · Working and answer

Working and answer

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## 4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

**9** A geometric series $G$ has common ratio $r$ where $r > 0$

The third term of $G$ is $\frac{27}{2}$ and the sum of the first three terms of $G$ is $\frac{57}{2}$

Given that the sum to $n$ terms of $G$ is $S_n$

(a) show that $S_n = \sum_{j=1}^{n} 4\left(\frac{3}{2}\right)^j$

**(8)**

Given that $S_k > 50\,000$

(b) show that the least value of $k$ is given by

$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

**(3)**

(c) Hence find the least value of $k$

**(1)**

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## 4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

**Question 9 continued**

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## 4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

**Question 9 continued**

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## 4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.



## 4. Question 9 · part c · 1 mark · Working and answer

Working and answer

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## 5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

**7** A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of $x$.

**(5)**

Given that $x < 1$,

(b) show that the series is convergent.

**(2)**

The sum to infinity of the series is $S$.

(c) Find the value of $S$.

**(2)**

The sum of the first $n$ terms of the series is $S_n$

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of $n$.

**(3)**

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## 5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

**Question 7 continued**

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## 5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

**Question 7 continued**

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## 5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.



**Question 7 continued**

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**(Total for Question 7 is 12 marks)**

## 5. Question 7 · part d · 3 marks · Working and answer

Working and answer

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## 6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**8** The $n$th term of a geometric series $G$ is $U_n$ and the sum of the first $n$ terms of $G$ is $S_n$

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of $U_5$

**(1)**

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where $A$ and $B$ are integers to be found.

**(3)**

The sum to infinity of $G$ is $S$

(c) Find the least value of $n$ such that $S - S_n < 0.045$

**(6)**

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## 6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 8 continued**

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## 6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 8 continued**

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## 6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

**Question 8 continued**

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**(Total for Question 8 is 10 marks)**

## 6. Question 8 · part c · 6 marks · Working and answer

Working and answer

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## 7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

**8** The sum of the first 2 terms of a geometric series $G$ is 360
The sum of the 2nd and 3rd terms of $G$ is 288

The $n$th term of $G$ is $U_n$

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where $A$ is an integer to be found.

(7)

(b) Explain why $G$ is convergent.

(1)

(c) Hence find the sum to infinity of $G$

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

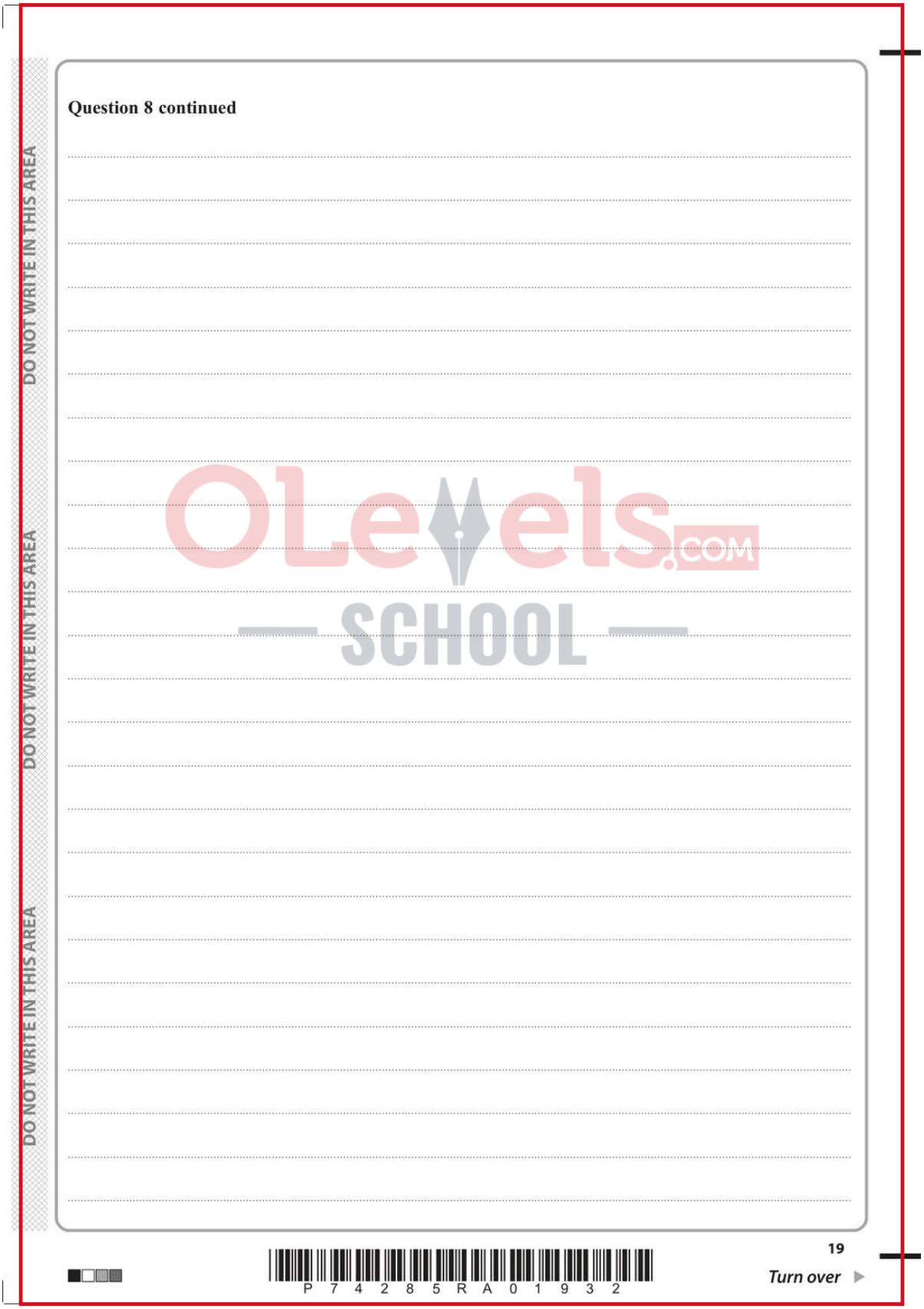
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## 7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.



**Question 8 continued**

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*Turn over*

## 7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

**Question 8 continued**

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## 7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

**Question 8 continued**

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**(Total for Question 8 is 14 marks)**

## 7. Question 8 · part d · 4 marks · Working and answer

Working and answer

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## 8. Question 6 · part b.ii · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b.ii only.

**6** (i) A geometric series $G$ has first term $\left(3+\sqrt{3}\right)$ and common ratio $r$

The second term of $G$ is $\sqrt{3}$

(a) Find, showing all your working, the exact value of $r$

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where $p$ and $q$ are integers to be found.

**(2)**

(b) Explain why $G$ is convergent.

**(1)**

(ii) A different geometric series $H$ has first term 8 and common ratio 0.6

The sum to $n$ terms of $H$ is $S_n$ and the sum to infinity of $H$ is $T$

Find, using logarithms, the least value of $n$ such that $T - S_n < 0.12$

**(5)**

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**12**

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## 8. Question 6 · part b.ii · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b.ii only.

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**Question 6 continued**

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**(Total for Question 6 is 8 marks)**

## 8. Question 6 · part b.ii · 5 marks · Working and answer

Working and answer

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## 9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

**7** The geometric series $G$ has first term $a$, common ratio $r$ and $n$th term $u_n$

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find $a$ in terms of e and $x$.

(3)

Given that the sum to infinity of $G$ can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for $p$ in terms of $x$.

(3)

Given that $u_{18} > 1.6$ and that $x$ is an integer,

(d) find the least value of $x$.

(4)

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## 9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

**Question 7 continued**

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## 9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

**Question 7 continued**

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## 9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

**Question 7 continued**

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**(Total for Question 7 is 13 marks)**

## 9. Question 7 · part d · 4 marks · Working and answer

Working and answer

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## 10. Question 7 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

**7** A geometric series $G$ has first term $a$ and common ratio $r$

The sum of the first three terms of $G$ is $\frac{61}{6}$

The sum to infinity of $G$ is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of $a$

**(6)**

The sum of the first $n$ terms of $G$ is $S_n$

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

**(2)**

(c) Hence find the least value of $n$

**(2)**

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## 10. Question 7 · part b · 2 marks · Working and answer

Working and answer

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## 11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

**7** (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

**(2)**

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

**(4)**

A geometric series $G$ has first term $a$ and common ratio $r$
The third term of $G$ is 9 and the sum to infinity of $G$ is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

**(3)**

Given that $r$ is a rational number

(d) write down the value of $r$

**(1)**

(e) show that $a = 144$

**(2)**

The sum to $n$ terms of $G$ is $S_n$

(f) Using logarithms, find the least value of $n$ such that $S_n > 191.9$

**(4)**

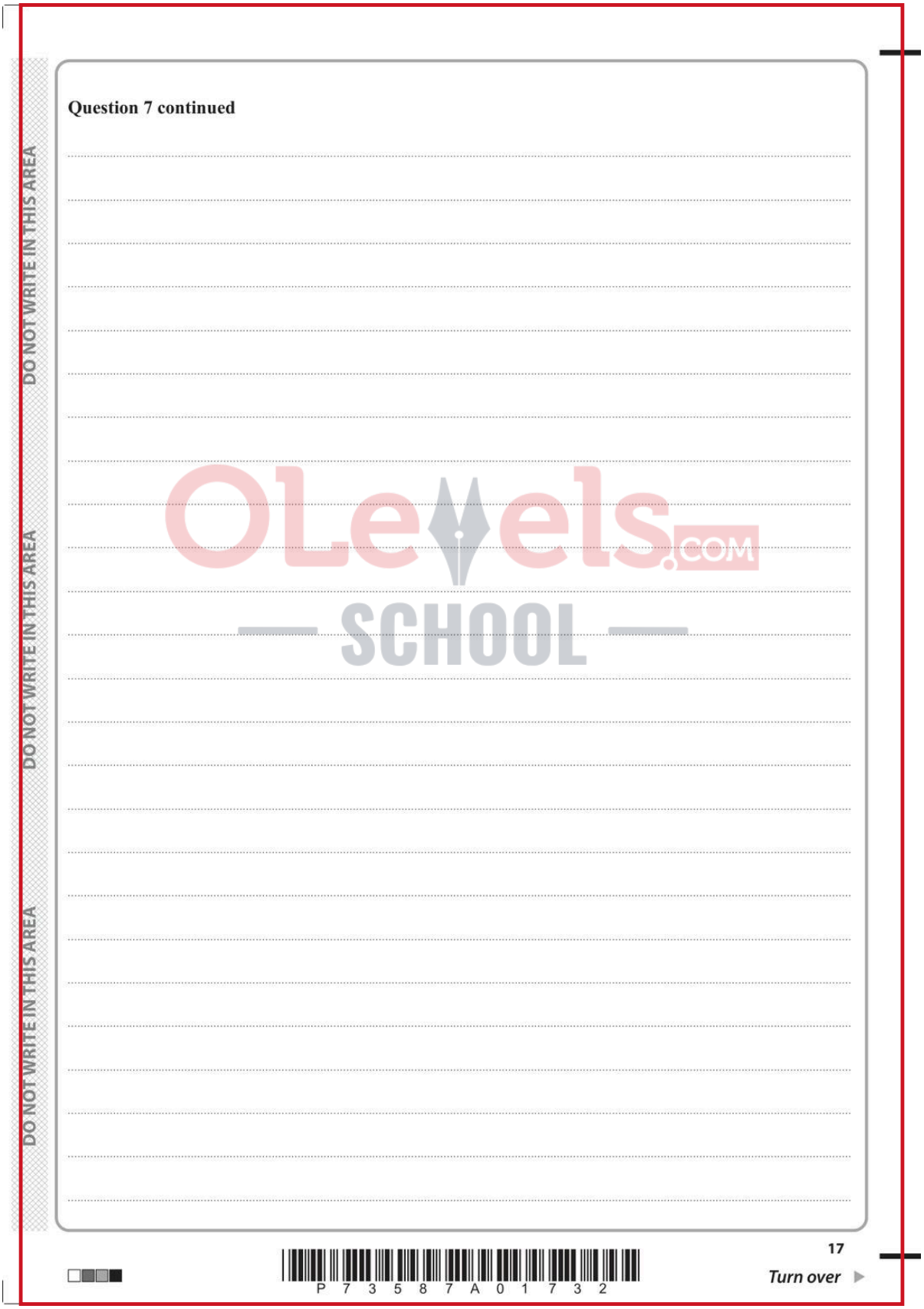
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## 11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.



**Question 7 continued**

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*Turn over*

## 11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

**Question 7 continued**

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## 11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

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**Question 7 continued**

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**(Total for Question 7 is 16 marks)**

## 11. Question 7 · part f · 4 marks · Working and answer

Working and answer

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## 12. Question 7 · part c · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

**7** A geometric series $G$ with common ratio $r$, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of $r$

**(2)**

Given that $r > 0$

(b) find the sum to infinity of $G$

**(2)**

The sum to $n$ terms of $G$ is greater than 33

(c) Find, using logarithms, the least possible value of $n$
Show your working clearly.

**(5)**

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## 12. Question 7 · part c · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

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**Question 7 continued**

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**(Total for Question 7 is 9 marks)**

## 12. Question 7 · part c · 5 marks · Working and answer

Working and answer

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## 13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

**7** The sum of the first $n$ terms of an arithmetic series is $A_n$ where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

**(2)**

The sum of the first $n$ terms of a geometric series is $G_n$ where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

**(2)**

(c) Find the value of $n$ for which $A_{14} - 6 = G_n$

**(5)**

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## 13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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## 13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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## 13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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**(Total for Question 7 is 9 marks)**

## 13. Question 7 · part c · 5 marks · Working and answer

Working and answer

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## 14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

**7** A geometric series $G$ has first term $a$ and common ratio $r$

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(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of $a$

**(6)**

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Given that $S_n > 19.8$

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**(2)**

(c) Hence find the least value of $n$

**(2)**

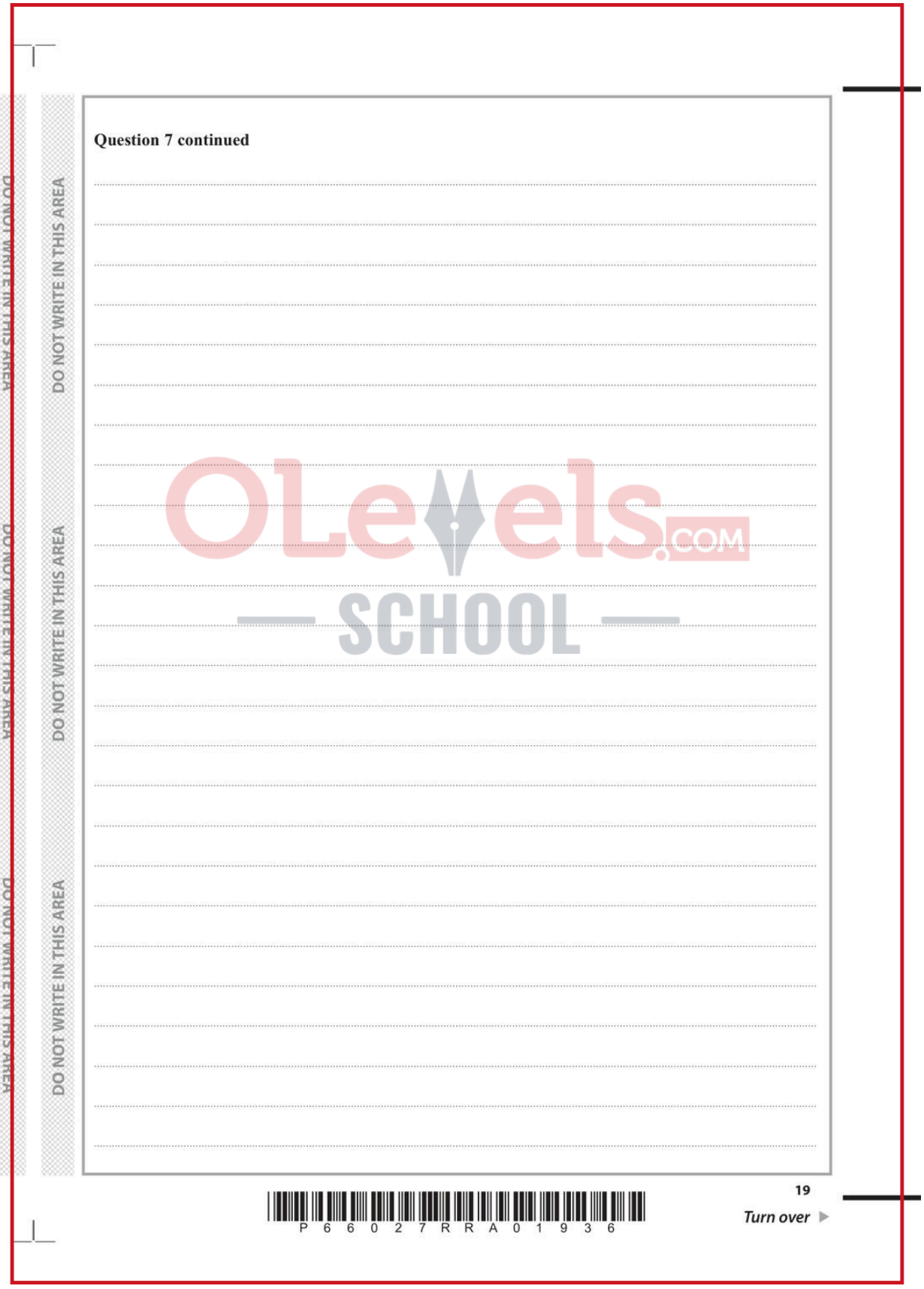
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## 14. Question 7 · part c · 2 marks

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Further practice · Attempt part c only.



## 14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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## 14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

**Question 7 continued**

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**(Total for Question 7 is 10 marks)**

## 14. Question 7 · part c · 2 marks · Working and answer

Working and answer

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